Do Now · retrieval

Operating systems and assembly challenge

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which OS component queues print requests?

Q2

What does LMC INP do to ACC?

Q3

What does Waiting mean?

Q4

Rising-edge D flip-flop has Q=1 and D=0 immediately before the active edge. What is Q after?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — LMC INP overwrites the accumulator; STA saves the value to a mailbox.

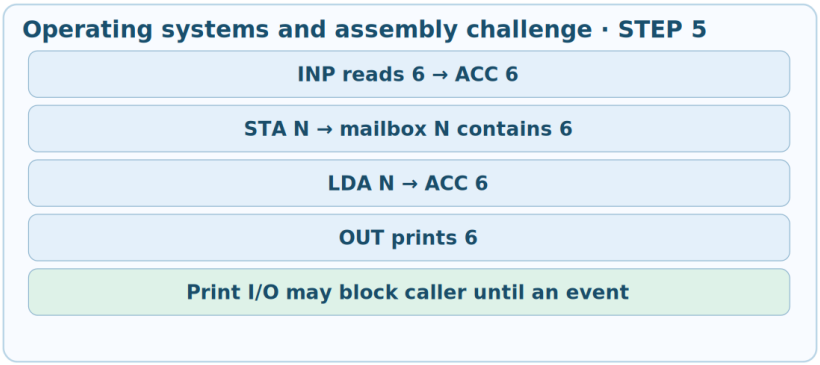
Core idea 2

Trace — Processes change between Ready, Running and Waiting as the OS schedules and responds to I/O.

Core idea 3

Explain — Spooling enables multiple programs to submit print jobs without monopolising the printer.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

ACC — Accumulator

Ready — Eligible for CPU

Running — Using CPU

Waiting — Blocked on event

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

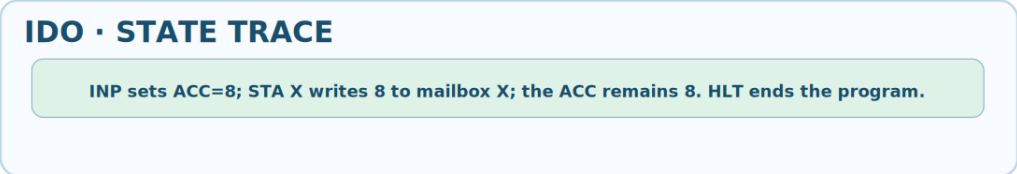
Problem

Trace an LMC fragment with input 8.

Follow the worked method

1. Predict first — INP sets ACC=8; STA X writes 8 to mailbox X; the ACC remains 8. HLT ends the program.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

The process waits for disk completion, then becomes Ready for selection by the scheduler.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State ACC and X after input 8 followed by INP; STA X. [4]

Question 2

Describe Ready, Running, Waiting and return to Ready during a disk read. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Describe OS file access, I/O process state and spooling when an app prints saved data. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A blocked process continues using the CPU until its I/O completes.

Use key vocabulary from memory

1. Explain ACC in your own words.

2. Explain Ready in your own words.

3. Explain Running in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.